# Mariano Rodriguez

**Chief Scientific Officer | Applied AI Research Leader**

PhD Computer Vision & Machine Learning | Speech, Vision & Language Models in Production

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> Applied-AI research leader who converts research into shipped, revenue-bearing product. Co-founder and CSO of **Freeda**, where I owned the scientific strategy and AI roadmap and led a team of 3 ML engineers building AI compliance review for the construction industry. Previously founding ML engineer at **Gladia** (speech-to-text API, PAYG and enterprise customers; later acquired by OVH), ML engineer at **Powder** (highlight detection in gameplay video), and sole technical founder of **Nano Quant Services**, a value-stock ML product I built and ran from zero. PhD and postdoc in computer vision from ENS Paris-Saclay; author at WACV, CVPR, ICIP and SIIMS; 6 years turning applied research into systems that hold up under production load and customer scrutiny.

## Selected Impact

- **Built a research function from scratch.** Hired and led 3 ML engineers at Freeda; set the AI roadmap, the evaluation bar and the in-house-vs-foundation-model boundary.
- **Research that reached the P&L.** Compliance workflows, engine and infrastructure delivering the fastest verifications in the market, supporting a EUR 1M+ book of business, and underpinning a EUR 2.5M round closed with Frst and Brick & Mortar plus EUR 20M with Notion as lead investor.
- **Every layer of Freeda's AI stack shipped under my direction.** Perception models beating vendor baselines, plan description layer, agentic verification, document intelligence, codegen mini-apps, and self-hosted Triton/Celery infra.
- **Inference economics at scale.** As founding ML engineer at Gladia, rebuilt Whisper serving to lower cost per hour of audio at better WER than the market at the time.
- **Zero-to-one, with capital discipline.** Sole technical founder of Nano Quant Services: full product — scraping, training, assistant with hybrid search & contextual RAG, backend + frontend — shipped in less than 6 months.

## Leadership & Experience

### Co-Founder & Chief Scientific Officer
**Nov 2024 – Nov 2026**

**Freeda, Paris**

*Owned scientific strategy, research roadmap and the production AI platform for automated compliance review of pre-construction plans against building regulation.*

- **Team & roadmap.** Built and led a team of 3 ML engineers; owned hiring and the AI roadmap.
- **Product impact.** Took the compliance workflows, engine and infrastructure from blank page to production in a few months, cutting manual plan-review time 3× with respect to the market standards.
- **Commercial results.** Live with 10+ paying customers within 10 months; the scientific work was the direct technical basis of the EUR 3.5M round raised from Frst and Brick & Mortar, and EUR 18.5M from Notion.
- **Cost & reliability.** Architected the self-hosted serving stack (GCP Compute Engine, NVIDIA Triton, Celery, LangChain, 4 DBs [Elasticsearch, Postgres, Redis, ChromaDB], monitoring [Grafana, Loki, Prometheus], GitHub CI/CD, auto-deploys); the first working prototype ran at < EUR 300/month with an NVIDIA L4 GPU.

### Founder & Chief Technology Officer
**May 2024 – Nov 2024**

**Nano Quant Services, Paris**

*Sole technical founder. Built and operated a complete machine-learning product for identifying value stocks, then made the call on its future.*

- **Built from zero.** Shipped an end-to-end product alone in 6 months: data acquisition (Selenium), model training, RAG retrieval, FastAPI backend, React/Node frontend, self-managed Kubernetes deployment.
- **Capital discipline.** Reached 2 paying users on EUR 20/month of infrastructure, zero outside capital and 4 months of runway — no outside engineering spend.

### Founding Machine Learning Engineer — Speech
**Jan 2023 – May 2024**

**Gladia, Paris**

*Applied research at commercial scale: speech-to-text infrastructure sold as a public API to 100+ customers (PAYG and enterprise contracts); Gladia was later acquired by OVH.*

- **Inference research with a direct margin effect.** Rebuilt Whisper for production serving: lower cost per hour of audio at better WER than the market at the time.
- **Owned a research area end to end.** Speaker diarization and voice-activity detection (pyannote, silero), which unlocked a customer segment: the meeting-transcription use case.

### Machine Learning Engineer — Video & Multimodal
**Jul 2021 – Jan 2023**

**Powder, Paris**

*Applied computer vision inside a consumer gaming product.*

- **Research shipped to consumers.** Event and highlight detection in gameplay streamer video.
- **On-device deployment.** Game identification in video and spoken-language identification exported to ONNX/CoreML.
- **State-of-the-art R&D.** Multi-annotation attention model for video summarization, presented at CVPR Workshop 2023.

### Postdoctoral Researcher, Applied Mathematics
**Jul 2020 – Jul 2021**

**École Normale Supérieure Paris-Saclay**

*Industry-funded research contracts — research run to an external client's specification and deadline. Advisor: Gabriele Facciolo.*

- **Huawei.** Recurrent-CNN joint video denoising and demosaicing delivered under contract; published at WACV 2023.
- **Kayrros.** CNN cloud detectors for satellite imagery.

## Technical Depth

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|---|---|
| **Research** | Computer vision, NLP, speech recognition & diarization, agents/skills/RAG, video understanding, geometric estimation, optimisation, applied mathematics |
| **Modelling** | PyTorch, PyTorch Lightning, TensorFlow, Keras, OpenCV, scikit-learn, LangChain, ONNX, CoreML |
| **Serving** | GCP, AWS, Slurm, Kubernetes, Docker Swarm, NVIDIA Triton Inference Servers, FastAPI, Flask, flame-graph profiling |
| **Engineering** | Python, C++, ctypes (C++↔Python) and MEX (C++↔MATLAB) bindings, PyTest (fixtures & mocks), SQLAlchemy/Postgres, React & Node, GitHub CI/CD, Cookiecutter, Task, Make & CMake, Unix, LaTeX/TikZ |
| **Scientific tooling** | Comet/WandB, Gin config, QGIS, MATLAB |
| **Languages** | Spanish (native), French (fluent), English (fluent) |

## Education

**PhD, Computer Vision & Machine Learning** — École Normale Supérieure Paris-Saclay. Advisors: Jean-Michel Morel & Julie Delon. 2016 – 2020

**MSc "Mathématiques, Vision, Apprentissage" (MVA)** — ENS Paris-Saclay. Highest Honors. 2015 – 2016

**MSc 1 "Mathématiques et Modélisation"** — Université Paris Cité. Highest Honors. 2014 – 2015

**BSc Mathematics** — Universidad de La Habana. Highest Honors; top student of the faculty. 2007 – 2011

## Selected Publications

**Video joint denoising and demosaicing with recurrent CNNs** WACV 2023

**Multi-annotation attention model for video summarization** CVPR-W 2023

**Robust estimation of local affine maps and its applications to image matching** WACV 2020

**Covering the Space of Tilts** SIIMS 2018

Full list: rdguez-mariano.github.io/publications

## Awards & Fellowships

**Laureate, FMJH Doctoral Program** — Fondation Mathématique Jacques Hadamard. Three-year competitive fellowship. 2016

**Laureate, PGSM Masters Scholarship** — Paris Graduate School of Mathematics. Two-year international scholarship. 2014

**Top student, Faculty of Mathematics & Computer Science** — Awarded by the vice-chancellor, University of Havana. 2011